

CHAMBERLAIN, SD, USA

WMO: 726530

Lat: 43.767N

Lon: 99.318W

Elev: 519

StdP: 95.24

Time Zone: -6.00 (NAC)

Period: 03-19

WBAN: 94943

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.3	-19.5	-26.0	0.4	-21.7	-23.5	0.5	-19.2	13.9	-5.6	12.4	-5.9	3.6	280	0.640

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	35.9	22.4	33.6	22.0	31.7	21.5	25.3	30.8	24.2	29.9	23.2	29.0	5.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	20.0	27.9	22.7	18.5	27.1	21.4	17.2	26.4	80.6	30.7	76.0	29.9	71.7	28.7	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.5	11.2	9.9	DB	-27.4	39.6	3.0	3.0	-29.5	41.7	-31.2	43.4	-32.9	45.1	-35.0	47.2	
			WB	-27.6	26.5	2.9	1.4	-29.7	27.5	-31.4	28.3	-33.0	29.1	-35.1	30.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	-6.1	-5.1	2.6	8.7	14.7	20.8	24.6	22.8	18.3	10.1	2.6	-4.7	(6)
	DBStd	12.14	7.37	7.19	7.23	5.69	4.89	3.72	3.63	3.42	4.64	5.41	6.10	6.50	(7)
	HDD10.0	2023	498	423	246	88	14	0	0	0	2	66	231	457	(8)
	HDD18.3	3856	756	656	488	291	134	19	2	5	57	259	473	715	(9)
	CDD10.0	1724	0	1	16	49	158	323	452	398	250	69	8	0	(10)
	CDD18.3	515	0	0	1	2	20	92	196	144	55	5	0	0	(11)
	CDH23.3	5614	0	0	14	56	257	931	2212	1430	628	84	2	0	(12)
	CDH26.7	2414	0	0	2	13	83	344	1088	621	241	22	0	0	(13)
(14) Wind	WSAvg	4.7	4.6	4.6	5.1	5.4	5.1	4.5	4.1	4.1	4.6	4.7	4.4	4.5	(14)
(15) Precipitation	PrecAvg	562	11	15	25	63	86	111	61	64	54	46	18	10	(15)
	PrecMax	742	42	31	60	160	162	209	146	159	172	164	46	38	(16)
	PrecMin	362	1	1	0	13	13	43	4	3	10	0	0	0	(17)
	PrecStd	108	9	8	17	36	46	42	40	35	38	44	14	10	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	16.8	25.7	28.6	32.4	34.8	40.8	37.7	35.5	29.7	22.4	13.5	(19)
		MCWB	6.3	8.9	13.6	15.0	19.8	22.5	22.1	22.2	20.4	16.5	11.8	6.3	(20)
	2%	DB	8.7	11.2	20.7	24.4	28.9	32.5	37.1	34.3	32.3	25.7	17.6	9.2	(21)
		MCWB	3.9	5.7	11.3	14.1	18.6	22.0	22.8	22.0	20.5	15.6	9.7	4.2	(22)
	5%	DB	6.0	7.5	17.0	21.3	26.2	30.5	34.4	32.2	29.5	22.3	14.3	6.2	(23)
		MCWB	2.6	3.8	10.0	12.4	17.3	21.2	22.7	21.6	19.2	14.0	7.9	2.6	(24)
	10%	DB	3.7	4.3	13.5	18.5	23.4	28.5	32.2	30.1	27.2	19.2	11.4	3.6	(25)
		MCWB	1.3	1.6	7.8	11.0	16.2	20.1	22.3	21.1	18.4	12.2	6.4	1.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	9.7	15.2	16.9	22.0	26.9	27.4	25.7	24.0	18.7	12.8	7.0	(27)
		MCDB	12.7	16.4	19.6	24.6	29.3	31.4	32.1	30.9	28.4	26.5	20.0	12.0	(28)
	2%	WB	4.1	6.1	12.3	15.0	20.1	24.4	25.7	24.5	22.5	16.6	10.5	4.7	(29)
		MCDB	7.6	10.7	18.8	22.8	26.7	29.3	30.9	29.9	27.9	23.1	16.9	9.0	(30)
	5%	WB	2.8	3.7	10.2	13.2	18.6	22.6	24.7	23.6	21.1	14.7	8.4	2.9	(31)
		MCDB	5.6	7.4	16.6	19.6	24.5	27.9	30.6	29.0	27.0	21.4	14.0	5.8	(32)
	10%	WB	1.4	1.9	7.9	11.7	16.9	21.3	23.6	22.4	19.8	12.8	6.5	1.4	(33)
		MCDB	3.6	4.3	13.4	17.4	22.1	26.8	29.9	28.0	25.3	18.2	11.0	3.5	(34)
(35) Mean Daily Temperature Range	MDBR		10.2	10.8	11.8	12.2	12.0	12.1	13.2	12.6	13.3	12.4	11.7	9.5	(35)
	5% DB	MCDBR	13.3	14.7	18.3	18.0	15.9	14.7	15.9	15.6	16.9	17.5	16.5	13.3	(36)
		MCWBR	9.6	9.9	10.3	9.4	7.8	6.6	5.6	6.0	7.1	9.0	9.6	9.2	(37)
	5% WB	MCDBR	12.7	14.2	17.0	15.9	13.7	12.4	13.1	12.5	13.6	15.7	16.1	12.7	(38)
		MCWBR	9.5	9.6	9.9	9.0	7.5	6.8	6.1	6.3	6.8	8.7	9.5	9.1	(39)
(40) Clear-Sky Solar Irradiance	taub		0.263	0.279	0.304	0.344	0.366	0.377	0.392	0.392	0.350	0.316	0.277	0.263	(40)
	taud		2.498	2.474	2.477	2.360	2.382	2.391	2.374	2.356	2.444	2.509	2.533	2.494	(41)
	Ebn at Noon		888	928	942	920	905	892	874	862	877	870	862	851	(42)
	Edh at Noon		68	83	95	116	118	117	118	116	98	80	66	62	(43)
(44) All-Sky Solar Radiation	RadAvg		1.89	2.88	3.89	4.92	5.83	6.59	6.82	5.72	4.58	3.07	2.07	1.54	(44)
	RadStd		0.15	0.21	0.34	0.36	0.53	0.46	0.24	0.45	0.31	0.37	0.20	0.12	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page